



PRODUCT NAME

Industrial-grade Hand Sanitizer Gel Formula

Description

Alcohol-based hand sanitizer gel

Technical Specifications

pH Level:	6
Viscosity:	Medium
Shelf Life:	24 months
Batch Size:	100-500 kg
Processing Time:	2-4 hours
Temperature:	Room temperature
Storage Conditions:	Cool, dry place
Equipment:	Standard mixer
Certification:	Meets industry standards

Formulation Table

Sr.No	Ingredient	INCI Name	%	Function
1	Ethanol 96%	Alcohol	62.5%	Antimicrobial agent
2	Deionized Water	Water	26.04%	Diluent
3	Isopropyl Myristate	Isopropyl Myristate	2.08%	Emollient
4	Carbomer	Carbomer	0.52%	Thickening agent
5	Triethanolamine	Triethanolamine	0.42%	pH adjuster
6	Glycerin	Glycerin	5.21%	Humectant
7	Aloe Vera Extract	Aloe Barbadensis Leaf Juice	2.08%	Skin conditioning agent
8	Fragrance	Fragrance	0.1%	Fragrance
9	Preservative	Phenoxyethanol	0.52%	Preservative
10	Vitamin E	Tocopheryl Acetate	0.52%	Antioxidant
11	Colorant (optional)	Colorant	0.01%	Color
Total			100%	

Manufacturing Process

Phase 1: Manufacturing Phase

Phase 2: Manufacturing Phase

Phase 3: Manufacturing Phase

Phase 4: Manufacturing Phase

Phase 5: Manufacturing Phase

Phase 6: Manufacturing Phase

Phase 7: Manufacturing Phase

Phase 8: Manufacturing Phase

Required Equipment

Standard mixer

Safety Precautions

Handling: Wear appropriate PPE including gloves, safety glasses, lab coat.

PPE Requirements: Chemical-resistant gloves, safety goggles, protective clothing.

Storage: Store in cool, dry place away from direct sunlight. Keep containers tightly closed.

Storage Conditions: Cool, dry place

Packaging Notes

Packaging: Use chemically compatible containers (HDPE, glass, or PET).

Labeling: Include product name, ingredients, usage instructions, and safety warnings.

Certification: Meets industry standards

Scaling Note

Lab Scale: This formulation is designed for laboratory testing and development.

Pilot Scale: For pilot production, scale proportionally and verify all parameters.

Production Scale: Consider equipment limitations, mixing efficiency, and process validation.

Batch Size: Current formulation is optimized for 100-500 kg.

Scaling Factor: Maintain ingredient ratios while adjusting processing parameters as needed.

Product Usage Instructions & Application Guidelines

Apply as needed according to product instructions